TOM ALDRIDGE

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PERSONAL PROFILE

Motivated, highly adaptable software engineer with a longstanding passion for computers. Able to pick up and master new skills quickly with excellent communication and teamwork abilities. Hardworking, enthusiastic and a proven leader.

EDUCATION

University of Nottingham — *September 2018 – September 2022*
MSc Computational Neuroscience, Cognition and AI — Distinction, Top of Class
BSc Mathematics and Finance — 1^{st} Class (Hons)

Exeter School — *April 2012 – June 2018*
Maths, Further Maths, Economics, (Computer Science) A-Levels — ABA(A)
GCSEs — 8 A*, 3A

TECHNOLOGIES

Python: Flask, Gradio, Numpy, OpenCV, Pandas, Psycopg, PyTorch, Tensorflow, Whisper
Other languages: C++, HTML/CSS, Java, MATLAB, R, Rust, Swift
Other: AWS, Docker, Git, GitLab, Linux, REST API, SQL, Virtual Machines

DEGREE MODULES

Neural Computation — **Machine Learning in Science**

Computational Cognitive Psychology — **Practical Biomedical Modelling**

Scientific Computation and Numerical Analysis — **Optimisation**

Mathematical Finance — **Statistical Models and Methods**

85% dissertation, highest score across the school — **77% degree average, highest on course**

EXPERIENCE

Defence Science Technology Laboratory (dstl) — *October 2022 – Present*

- Led a team to AWS Deep Racer 2023, a reinforcement learning competition. Placed 1^{st}, 2^{nd}, 3^{rd} and 6^{th} in the 4 competitions available. Used AWS tech stack, reinforcement learning principles and reward functions to achieve top tier performance in simulation and on track.
- Developed an advanced synthetic data generation framework for use in a semi-supervised ML model involving complex custom image filters and augmentations, improving model precision from 88% to 94%.
- Converted fragmented ML inference code into GPU-enabled Docker containers for batch parallel processing on an NVIDIA DGX, multiplying processing speed by over 20x, alongside monitoring and maintaining the services.
- Implemented non-myopic sensor management algorithms for Stone Soup, dstl's open-source tracking and state estimation framework.
- Designed database scheme and instantiated PostgreSQL databases for storing hundreds of millions of rows of embedded images and metadata. Implemented Python SQL database connectors and wrote custom complex queries for querying 800 million rows.
- Developed a real-time speech-to-text application and API for low latency (<5s), high accuracy (word error rate < 5%) using OpenAI's Whisper as a backend.
- Produced well-documented and well-tested code using test-driven development, unit testing and integration testing, while adhering to PEP-8 standards, version control using Git, and engaging in frequent code reviews.

CFCLoanBot (https://github.com/tomaldridge12/loanbot) *March 2024 – Present*

- Developed a real-time football analytics Twitter/X bot for providing updates on Chelsea FC players who are not presently at the club (https://twitter.com/CFCLoanBot). Garnered over 1,000 unique followers in a single weekend.
- The multithreaded application periodically queries RestAPIs for in-match info and handles new events, such as goals, assists, substitutions and cards.
- A custom image generation pipeline automatically generates images alongside the tweets for increased visual interest.
- Future plans include adding monthly player reviews and integrating the Chelsea Women's team.

Blocfolio *November 2023 – Present*

- Solo developer of an indoor climbing-based social media platform built using Swift and SwiftUI for iOS.
- Users can upload photos, comment, interact, and share media with other users on the platform.
- Database, storage, and account authorisation implemented using Firebase.

Football Manager Scouting Platform *February 2023*

- Designed and created a local application for enabling data-led transfers in Football Manager, a popular football strategy game.
- Developed models that evaluate player performances based on in-game statistics and attributes using sci-kit learn and Numpy.
- Front-end delivered via Gradio to enable tweaking of search parameters and visualisation of results.

Nottingham Medical Imaging Academy *October 2021 – March 2022*

- Accepted into a program run by the Precision Imaging Beacon to provide training and experience in medical imaging. Delivered via 3 modules across the year:
- **Computing for Medical Imaging** covered installing and using key software tools for image analysis, including Git, Conda, Python and Jupyter.
- **Essential Medical Imaging Acquisition** discussed the main methods of medical imaging, focusing on how MRI imaging works, including image formation, pulse sequences, contrast imaging, fast imaging, and artefacts.
- **Essential Image Analysis** covered methods that are broadly applicable in medical imaging applications, such as viewing and manipulating medical images, registration, segmentation, regression and model fitting, classification and machine learning.

Portfolio Optimisation and Generation Project *Jan – April 2021 (Grade: First Class)*

- Tasked with creating software in order to create and automatically manage optimal stock portfolios with user defined risk and profitability characteristics.
- User was able to "buy" and "sell" individual stocks and add them to a cloud-based portfolio or use one of several bespoke algorithms to automatically generate a suitable portfolio. Automatic adjustments could be made by the software to sell unprofitable stocks and replace them with stocks that would suit the portfolio characteristics.
- Real-time stock data was imported from Yahoo Finance in order to generate covariance matrices for a shortlist of stocks that had desirable attributes. Constrained optimisation was performed to return the optimal stock weights within the portfolio, then time-series analysis was used to forecast the returns from the individual stocks.
- Project was delivered as a web app using R and various external packages such as RShiny and was accompanied by a 30-page report detailing the project, research, and results.

Minecraft Server and Plugin Development *2015 – 2017*

- Personally deployed and configured multiple custom game servers that enjoyed thousands of unique players over their lifespan, as well as creating the accompanying websites, online stores, and forums.
- Developed bespoke server plugins using Java in order to enhance the gaming experience for the players.